🧬 HEREDITY — FULL MIND MAP

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📑 Topics in this chapter:

1. Mendel — Father of Genetics & Laws of Inheritance

- Gregor Johann Mendel ko source "father of genetics / modern genetics" batata hai; laws of inheritance unhi se associated hain. *[Q1, Q2, Q3]*
- Mendel ne hybridization experiments garden pea par kiye aur pea ke seven types of qualities study karke Law of Dominance, Law of Segregation aur Law of Independent Assortment diye. *[Q4]*
- Mendel ka principle of inheritance sexual reproduction par based hai. *[Q5]*

2. Genome / Gene Mapping

- Gene mapping chromosome par gene ki location aur genes ke beech relative distance determine karne ki method hai. Genome mapping molecular markers ko genome par unki respective positions par place karta hai. *[Q6]*

3. Gene — Basic Unit of Heredity

- Gene heredity ki basic physical aur functional unit hai; genes DNA se bane genetic material ke functional units hain. *[Q7]*
- Source ke mutabik chromosome ke long coiled DNA molecule ke small parts ko gene kaha jata hai. *[Q8]*

4. Genetic Code — Codons & Main Features

- Genetic code genetic material ki information ko proteins mein translate karne ke rules ka set hai. DNA/mRNA nucleotides ko three-letter codons ke form mein read kiya jata hai; 64 codons 20 amino acids se correspond karte hain. *[Q9]*
- Source genetic code ko universal, triplet, commaless, non-overlapping, non-ambiguous aur redundant batata hai. AUG start codon hai; UAA, UAG aur UGA stop codons hain. *[Q9]*

> ⚠ **EXAM TRAP — Genetic Code Signals**
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> ▸ AUG = start codon; UAA, UAG, UGA = stop codons. Source Q9 mein all listed characteristics correct answer diye gaye hain. *[Q9]*

5. Mutation — Change in Gene Base Sequence

- Gene ke base sequence mein change ko mutation kaha jata hai. Source mutation ko gene structure ka heritable variant-producing change aur genetic variation ka ultimate source batata hai. *[Q10]*

6. Nucleus, Chromosomes & Hereditary Information

- Nucleus eukaryotic cell ka control centre aur genetic information repository hai; DNA ko house/replicate karke cell reproduction mein important role play karta hai. Isliye "nucleus does not play a role in cellular reproduction" statement source mein incorrect hai. *[Q11]*
- Hereditary characters descendants tak chromosomes ke through transfer hote hain. Source chromosomes ko nucleoprotein-made structures batata hai; "chromosome" term Waldeyer se associate karta hai aur genes chromosomes par located hote hain. *[Q12]*
- DNA living cells ke hereditary characters control/store/transfer karne wala predominant hereditary material hai; source RNA ko messenger aur kuch viruses mein hereditary material batata hai. *[Q13]*

7. DNA Complementary Base Pairing

- DNA mein specific base pairing A–T aur G–C hoti hai; dono strands complementary hote hain. Source isi complementary base pairing ko genetic information ko generation-to-generation store/transmit karne ke liye uniquely suited feature batata hai. *[Q14]*

8. Sexual Reproduction & Genetic Variation

- Sexual reproduction mein two parent cells contribute karte hain; source genetic variation ko blending of genes, chromosomal changes aur gene shuffling se associate karta hai. *[Q15]*

9. Chromosomal Theory of Inheritance

- Chromosomal theory of inheritance Walter Sutton (1902–03) aur Theodor Boveri ke independent work se associated hai. Source ke mutabik ye Mendelian inheritance ke mechanism ko chromosomes ke through explain karta hai. *[Q16]*

10. DNA Discovery vs Double-Helix Structure

- James D. Watson aur Francis Crick DNA ke double-helix structure/model (1953) se associated hain; source Maurice Wilkins ke saath 1962 Nobel Prize ka bhi zikr karta hai. *[Q17, Q18, Q19, Q20, Q21, Q22]*
- DNA molecule ki first discovery source Friedrich Miescher (1869) se associate karta hai; unhone ise "nuclein" kaha. *[Q23]*

> ⚠ **EXAM TRAP — "Who discovered DNA?"**
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> ▸ Watson + Crick = DNA double-helix structure/model; source Q23 ke mutabik DNA molecule first discovered by Friedrich Miescher (1869). *[Q17, Q18, Q19, Q20, Q21, Q22, Q23]*
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> ▸ Q23 ke options mein Miescher nahi tha, isliye source "None of the above" ko technically correct answer maanta hai. *[Q23]*

11. Har Gobind Khorana — Genetic Code & Artificial Gene

- Har Gobind Khorana, Robert W. Holley aur Marshall W. Nirenberg ko 1968 Nobel Prize genetic code interpretation aur protein synthesis mein DNA function ke work ke liye source mention karta hai. *[Q24]*
- Source Khorana ko laboratory mein artificial gene / short DNA sequence synthesize karne wala scientist batata hai. *[Q25]*

12. Jumping Genes — Barbara McClintock

- Barbara McClintock “jumping genes” / transposons ke principle se associated hain. Source unke maize heredity work aur 1983 Nobel Prize ka mention karta hai. *[Q26]*

13. DNA Components, Sugar & Ribozymes

- Source Q27 mein adenine, guanine, cytosine aur thymine ko DNA nucleobases batata hai; tyrosine amino acid hai aur DNA ka structural base nahi. *[Q27]*
- DNA mein deoxyribose sugar hoti hai; source ribose ko RNA ki sugar batata hai. *[Q28]*
- Ribozymes catalytic RNA molecules hain jo chemical reactions catalyze kar sakte hain; source ribosome mein amino acids ko protein chain mein join karne se bhi relate karta hai. *[Q29]*

14. Chromosome Number — Species Specificity

- Chromosome number species-to-species vary karta hai, lekin particular species ke liye constant rehta hai; source ke mutabik age ya weight ke saath normally change nahi hota. *[Q30]*

15. Sex Chromosomes & Human Karyotype

- Source many lizard species mein morphologically recognizable sex chromosomes absent hone aur environmental factors (jaise incubation temperature) se sex determination ka mention karta hai. *[Q31]*
- Human cells mein total 46 chromosomes = 23 pairs hote hain; 22 pairs autosomes aur 23rd pair sex chromosomes hota hai. *[Q32, Q33]*
- Females XX aur males XY hote hain. Source ke explanation ke mutabik human offspring ka sex male parent ke X- or Y-bearing sperm se determine hota hai: XX female, XY male. *[Q34, Q35, Q36]*

⚠ EXAM TRAP — Human Sex Determination

- ▸ Female = XX; Male = XY. Source Q34 mein Assertion “female plays major role” false aur Reason “women have two X chromosomes” true diya gaya hai. *[Q34]*
- ▸ Male child ke liye XY combination chahiye; Y chromosome male parent contribute karta hai. *[Q35, Q36]*

16. DNA Mutation Risk — Ionizing Radiation

- Ionizing radiation jaise X-rays/gamma rays DNA mutations ya permanent DNA changes cause kar sakte hain. Source Q37 mein repeated radiation exposure ki wajah se X-ray technician ko specified professionals mein higher risk answer karta hai. *[Q37]*